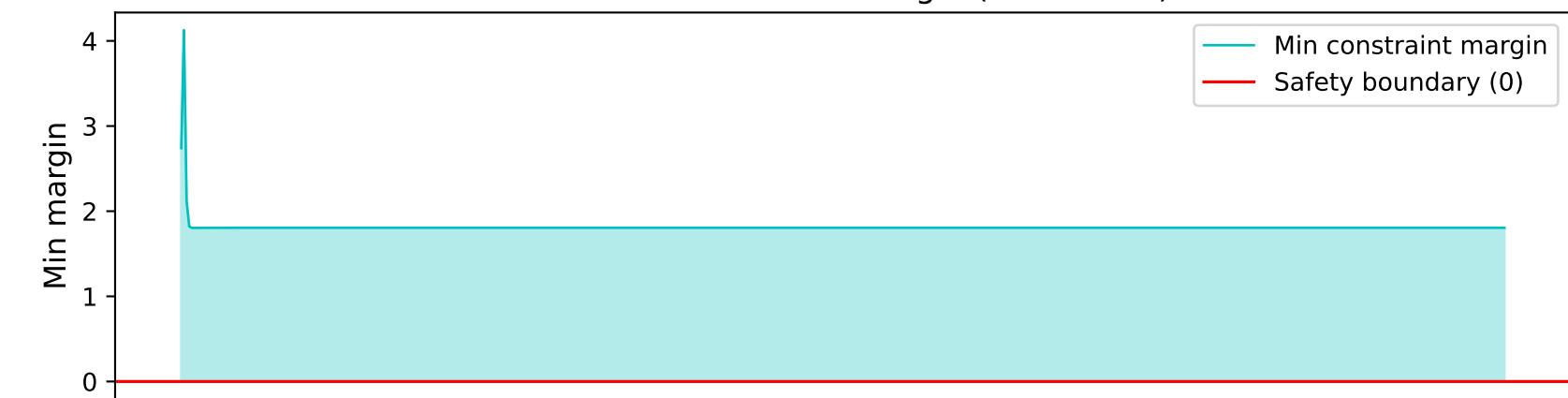


Safety Filter Behavior under S3: Coupled Perturbation (PPO-RHOCBF, $\kappa_\varepsilon = 1.0$)

Compositional Robustness Margin



Minimum Constraint Margin (safe if > 0)



QP Safety Filter Intervention

